

PREDICTING ICU PATIENT DETERIORATION AND MAPPING THEIR TRAJECTORIES

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Capstone project



WHY ICU DETERIORATION PREDICTION MATTERS

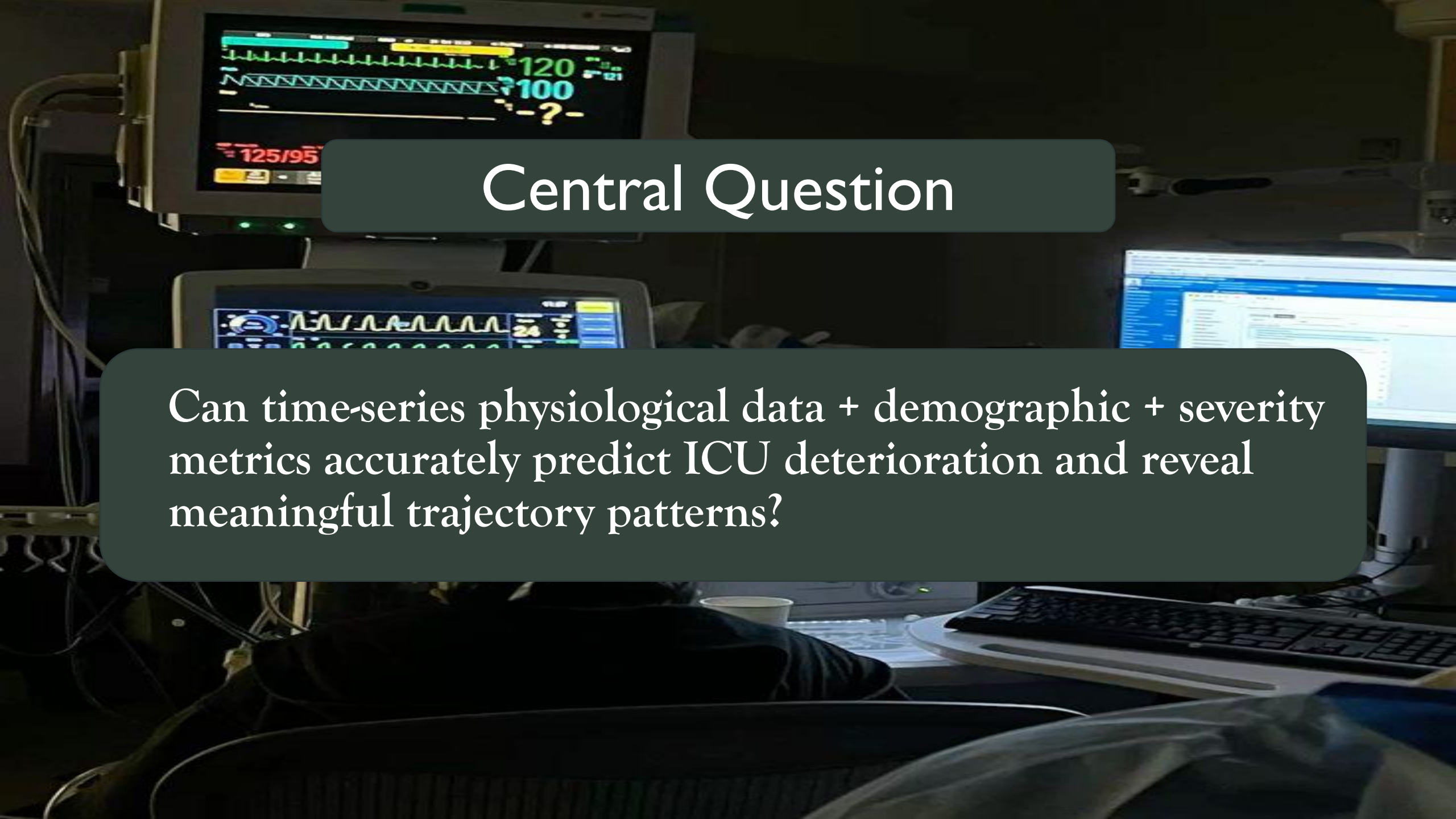
ICU patients can deteriorate rapidly

Early warning signs are subtle, nonlinear, and easily missed

Traditional scoring systems (e.g., APACHE) capture severity but not dynamics

Machine learning can detect complex, multi-system interactions



The background of the slide is a photograph of an Intensive Care Unit (ICU) monitoring room. It features several computer monitors displaying various physiological data. The top monitor shows a heart rate of 120, a respiratory rate of 100, and a blood pressure of 125/95. Other monitors in the background show additional waveforms and data. The room is dimly lit, with the primary light source being the screens.

Central Question

Can time-series physiological data + demographic + severity metrics accurately predict ICU deterioration and reveal meaningful trajectory patterns?



WHAT PRIOR WORK MISSED

- Small feature sets (mostly vital signs)
- Limited temporal modeling
- No clustering of patient trajectories
- Single-center datasets → limited generalizability

DATASET OVERVIEW

eICU Collaborative Research Database

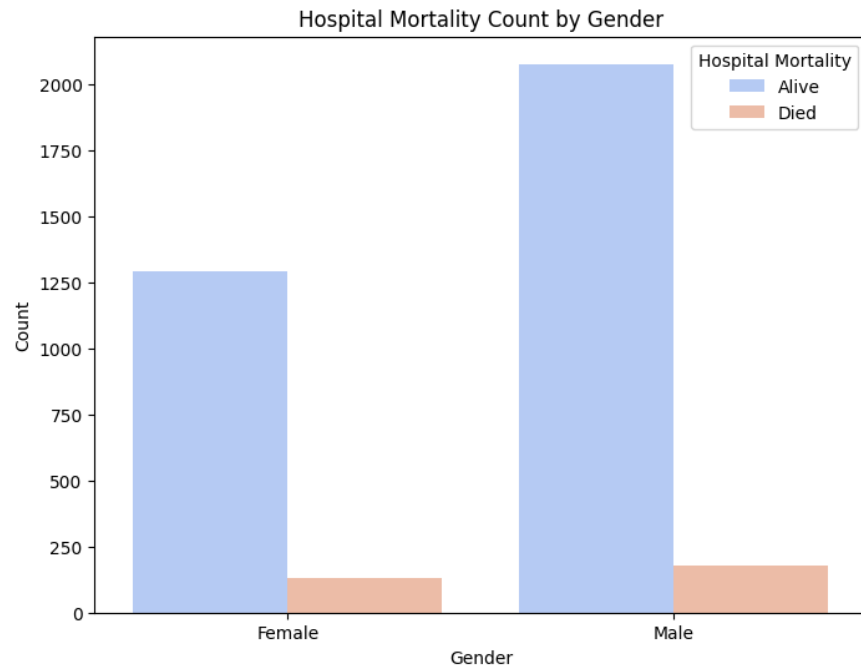
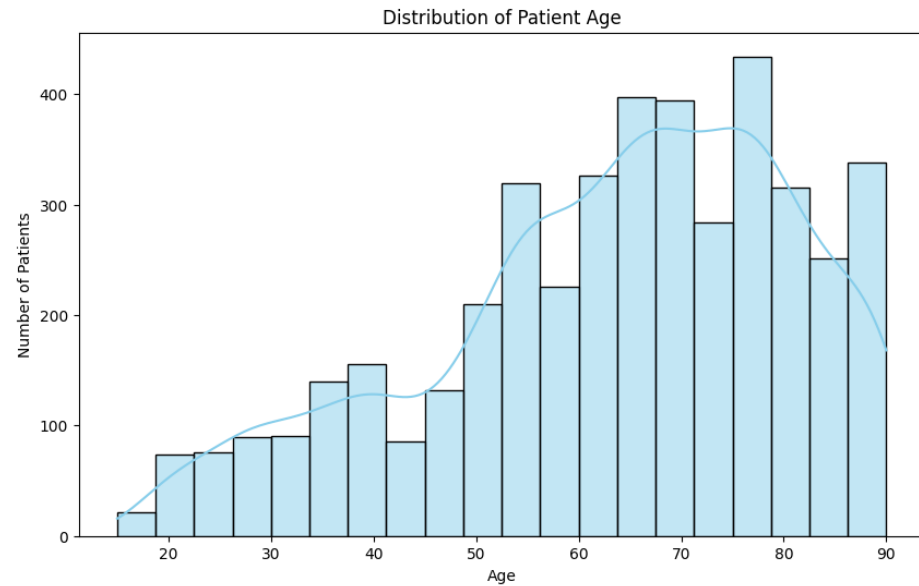
- Multi-hospital ICU dataset
- Demographics
- Vital signs (periodic + aperiodic)
- Labs
- Medications
- APACHE severity scores
- Mortality outcomes



DATA PREPROCESSING

- Standardized age, imputed missing values
- Cleaned labs, medications, diagnoses
- Forward/backward filled time-series vitals
- Merged all datasets into patient-level features



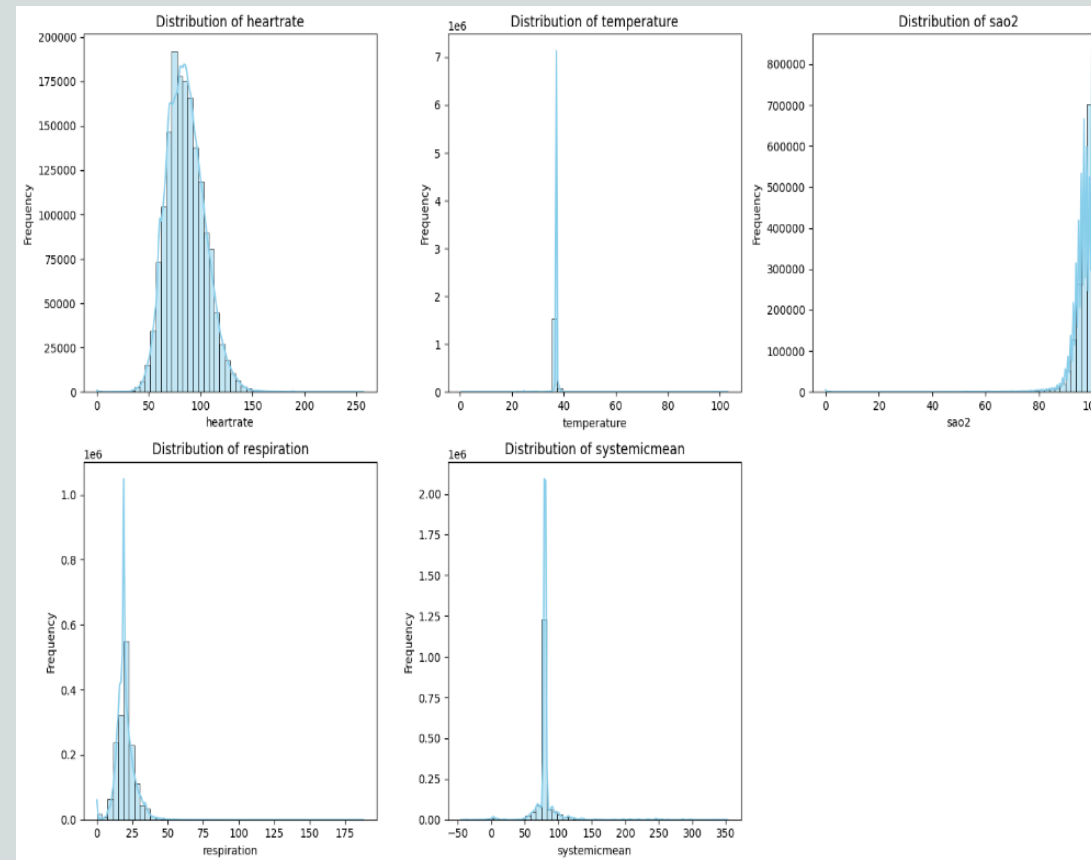
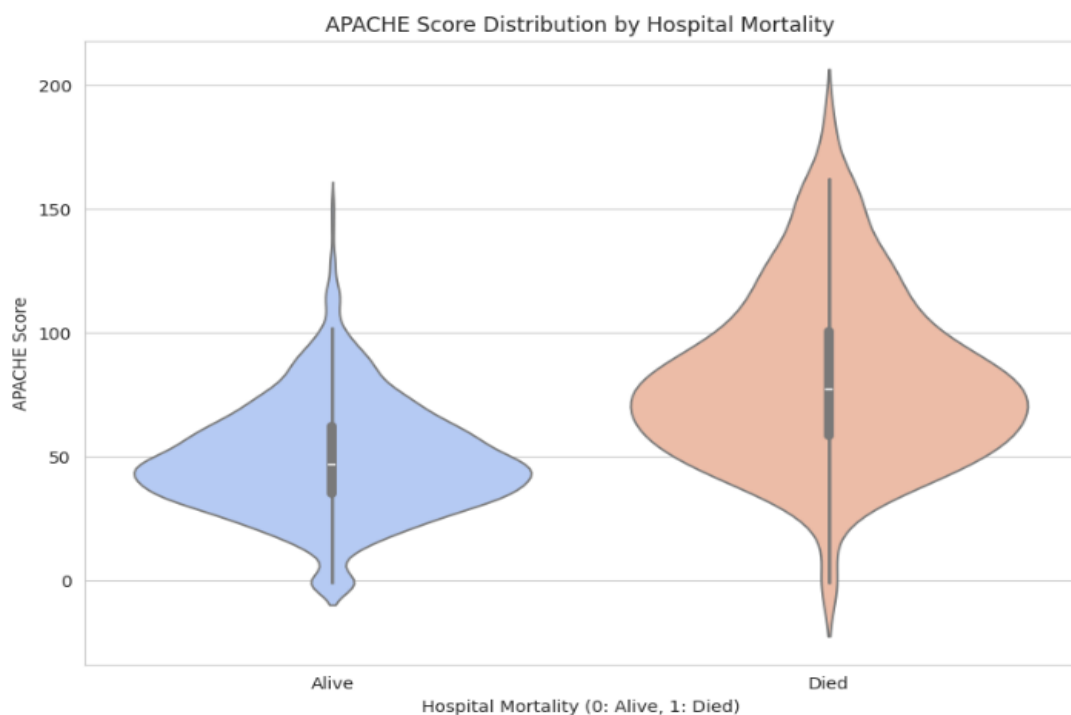


EXPLORATORY DATA ANALYSIS

Key EDA Findings

- Mortality similar across genders
- Age skewed older

EXPLORATORY DATA ANALYSIS



Key EDA Findings

- APACHE strongly correlated with mortality
- Vital signs show expected ranges + extreme values

Correlation Matrix of Key Numerical Features															
age	1.00	-0.09	-0.10	0.37	0.10	0.06	0.12	0.14	0.09	0.08	0.18	0.16	0.12	0.07	0.05
admissionheight	-0.09	1.00	0.25	-0.11	-0.09	-0.05	-0.02	-0.05	-0.03	-0.04	-0.03	-0.04	-0.04	-0.01	-0.01
admissionweight	-0.10	0.25	1.00	-0.07	-0.03	-0.03	-0.01	0.01	-0.00	-0.04	-0.05	-0.01	0.01	0.02	0.04
apachescore	0.37	-0.11	-0.07	1.00	0.95	0.57	0.32	0.67	0.22	0.46	0.36	0.43	0.23	0.31	0.51
physiologyscore	0.10	-0.09	-0.03	0.95	1.00	0.55	0.31	0.68	0.23	0.44	0.33	0.41	0.22	0.34	0.56
predictedmortality	0.06	-0.05	-0.03	0.57	0.55	1.00	0.21	0.63	0.11	0.69	0.23	0.42	0.08	0.18	0.36
actualmortality	0.12	-0.02	-0.01	0.32	0.31	0.21	1.00	0.22	0.06	0.16	0.75	0.11	-0.05	0.16	0.19
predictediculos	0.14	-0.05	0.01	0.67	0.68	0.63	0.22	1.00	0.31	0.38	0.26	0.61	0.35	0.43	0.74
actualiculos	0.09	-0.03	-0.00	0.22	0.23	0.11	0.06	0.31	1.00	0.08	0.10	0.19	0.55	0.71	0.30
predictedhospitalmortality	0.08	-0.04	-0.04	0.46	0.44	0.69	0.16	0.38	0.08	1.00	0.17	0.64	-0.07	0.14	0.25
actualhospitalmortality	0.18	-0.03	-0.05	0.36	0.33	0.23	0.75	0.26	0.10	0.17	1.00	0.13	0.01	0.19	0.22
predictedhospitallos	0.16	-0.04	-0.01	0.43	0.41	0.42	0.11	0.61	0.19	0.64	0.13	1.00	0.31	0.24	0.37
actualhospitallos	0.12	-0.04	0.01	0.23	0.22	0.08	-0.05	0.35	0.55	-0.07	0.01	0.31	1.00	0.41	0.27
actualventdays	0.07	-0.01	0.02	0.31	0.34	0.18	0.16	0.43	0.71	0.14	0.19	0.24	0.41	1.00	0.62
predventdays	0.05	-0.01	0.04	0.51	0.56	0.36	0.19	0.74	0.30	0.25	0.22	0.37	0.27	0.62	1.00
age	admissionheight	admissionweight	apachescore	acutephysiologyscore	predictedmortality	actualmortality	predictediculos	actualiculos	predictedhospitalmortality	actualhospitalmortality	predictedhospitallos	actualhospitallos	actualventdays	predventdays	

EXPLORATORY DATA ANALYSIS

- APACHE & acute physiology scores strongly correlate with mortality
- Ventilation days correlate with severity
- LOS correlates with predicted LOS
- Height/weight show weak correlations

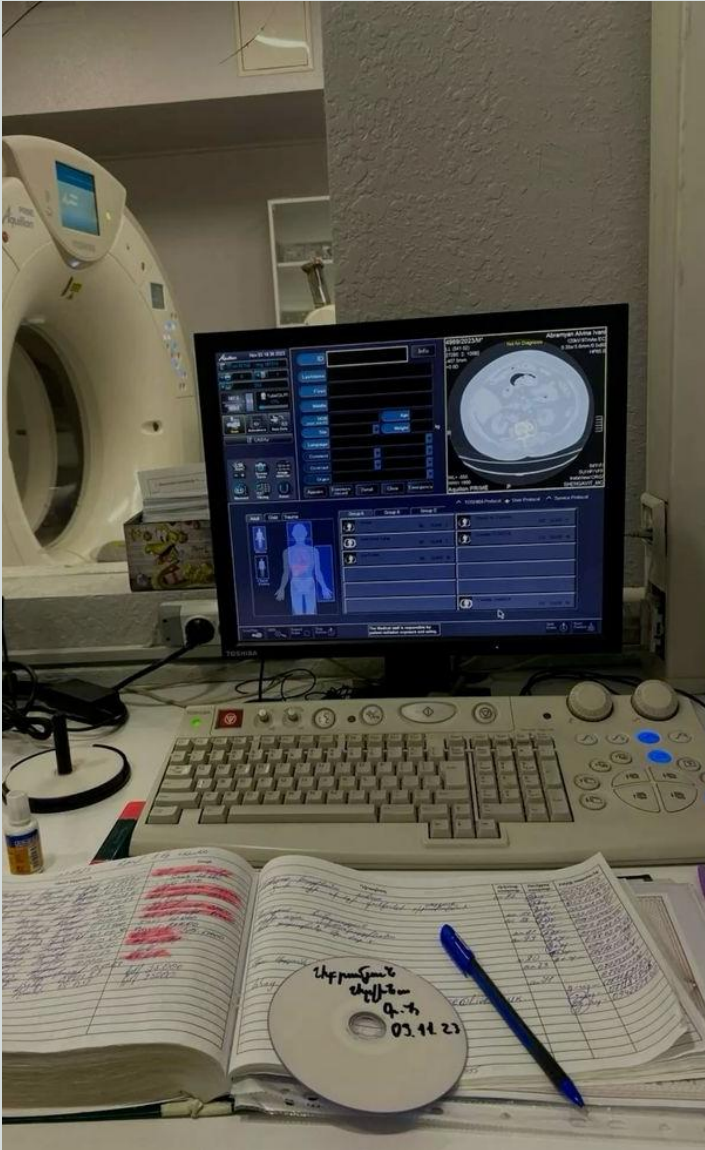


TIME-SERIES FEATURE ENGINEERING

- First 24 hours summarized
- Means, mins, maxes, SDs for vitals
- Lab summaries
- Medication counts + dosages
- Diagnosis/treatment counts
- One-hot encoding for categorical variables

PCA DIMENSIONALITY REDUCTION RESULTS

- 884 $\rightarrow$ 427 components (95% variance)
- PC1 = global physiological stability
- PC4 = respiratory function
- PC2/PC3 = renal + inflammatory patterns



SUPERVISED LEARNING OVERVIEW

Models Evaluated

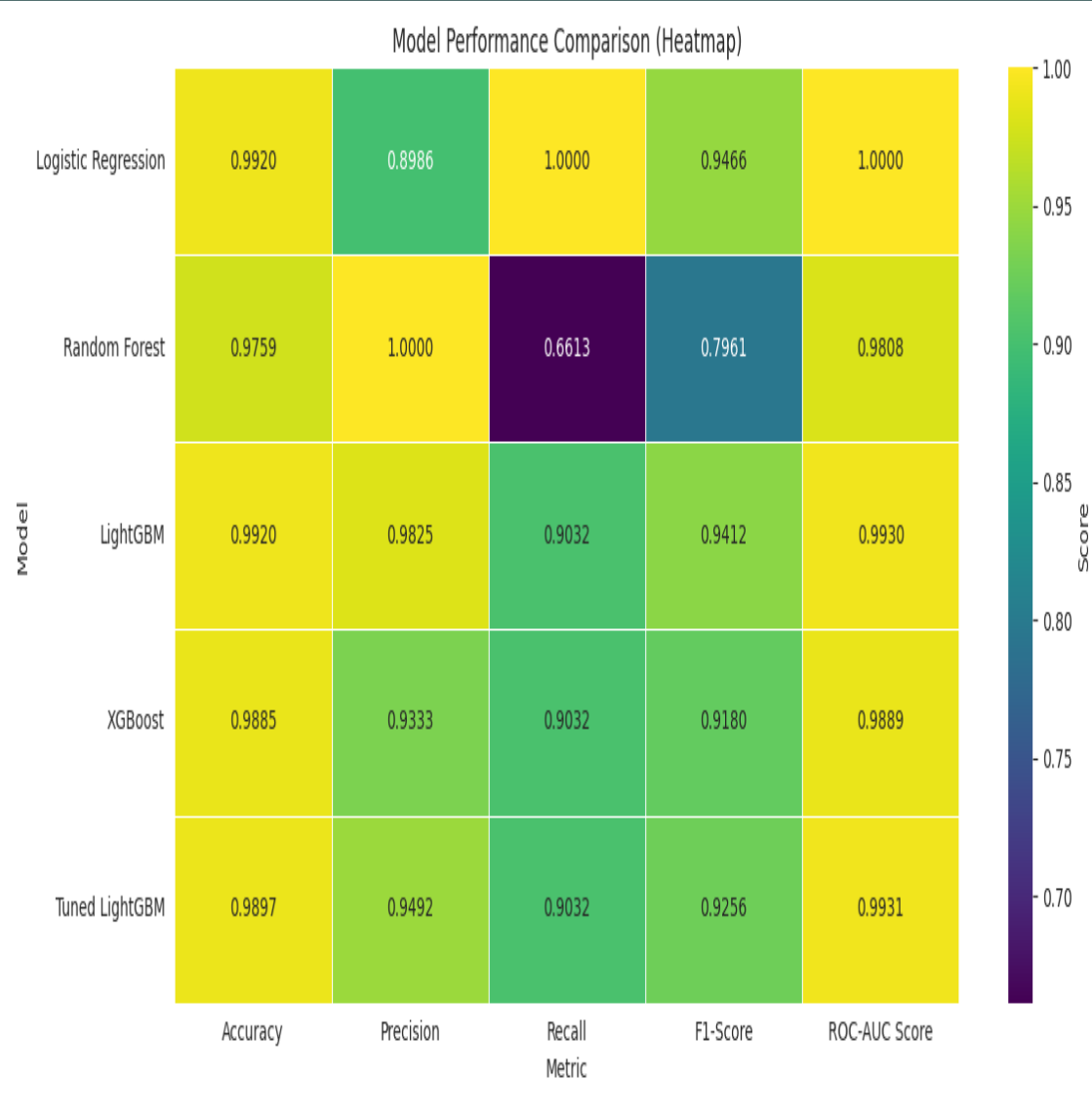
- Logistic Regression
- Random Forest
- LightGBM
- XGBoost
- KNN

Why This Set?

- Covers linear $\rightarrow$ nonlinear $\rightarrow$ boosting paradigms
- Allows comparison of interpretability vs. performance
- Handles class imbalance differently



SUPERVISED LEARNING OVERVIEW



Logistic Regression: perfect recall, lower precision

Random Forest: perfect precision, low recall

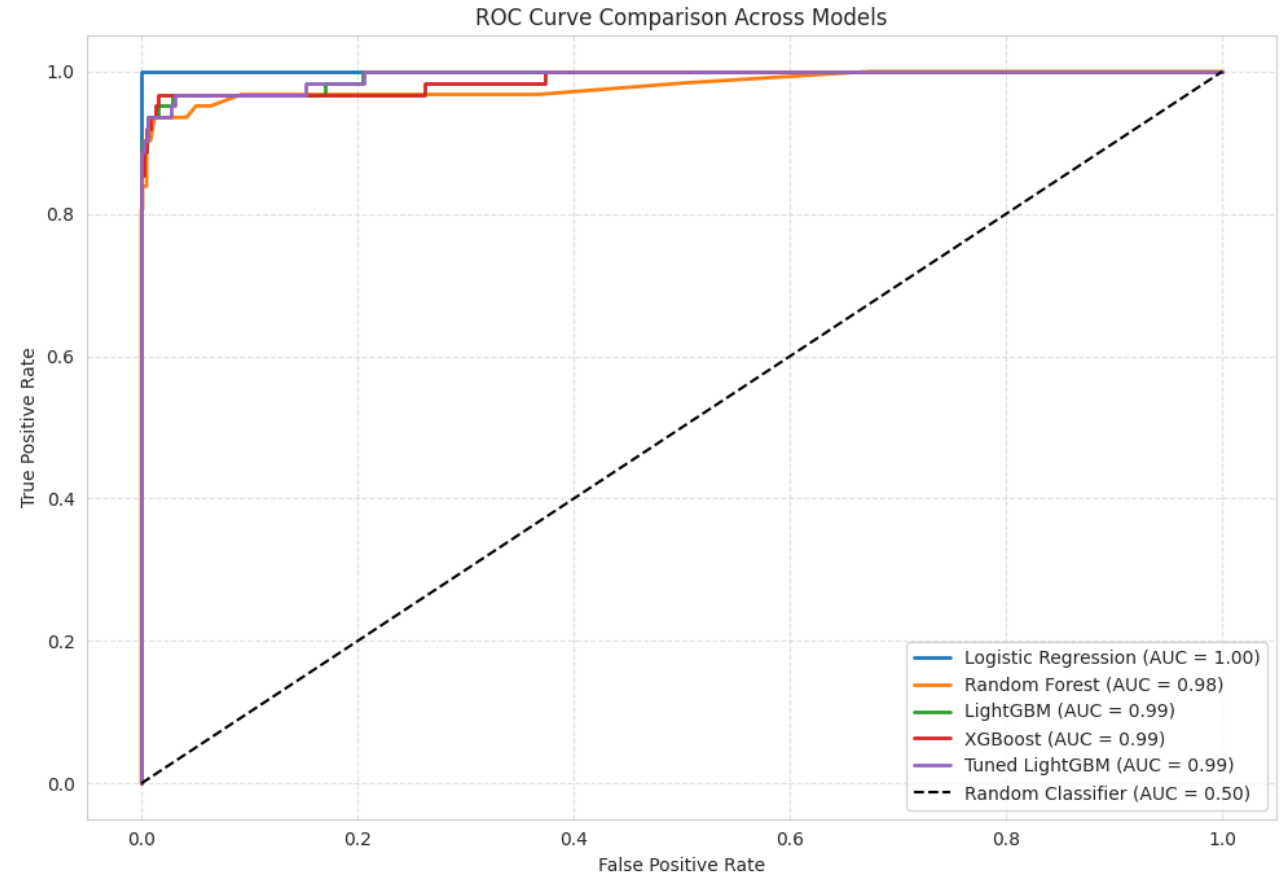
LightGBM: best balance across all metrics

XGBoost: strong but slightly lower precision

KNN: weakest due to high-dimensional space

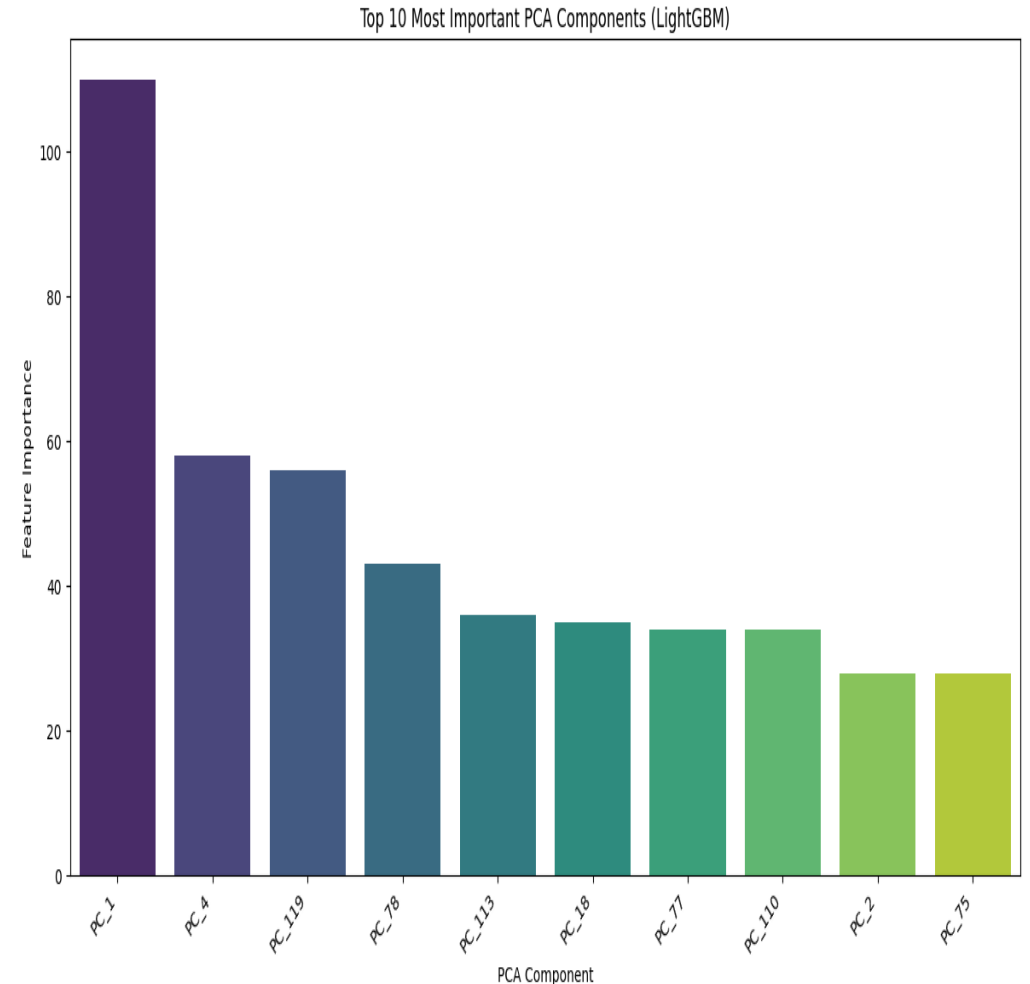
ROC CURVE INTERPRETATION

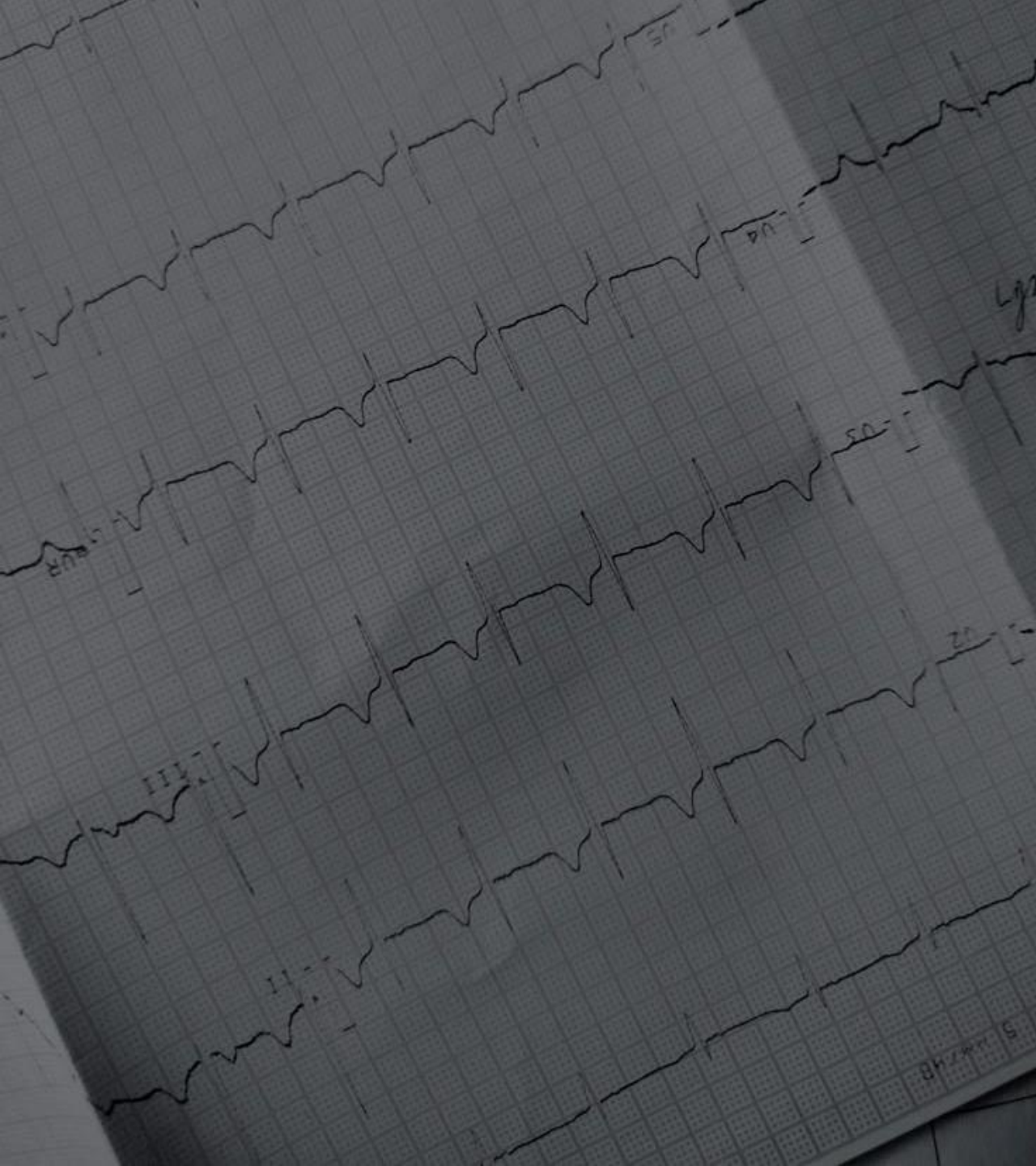
- **LightGBM:** $AUC = 0.9930 \rightarrow$ near-perfect discrimination
- **XGBoost:** $AUC = 0.9889 \rightarrow$ extremely strong performance
- **Logistic Regression:** $AUC = 1.0000 \rightarrow$ perfect separation on this dataset
- **Random Forest:** $AUC = 0.9808 \rightarrow$ strong but noticeably less sensitive at lower thresholds
- **KNN:** $AUC = 0.7961 \rightarrow$ poor discrimination compared to all other model



TOP 10 PCA COMPONENTS (LIGHTGBM IMPORTANCE)

- **PC1** contributes $\sim 40\text{--}45\%$ of total model importance $\rightarrow$ Dominant predictor capturing global physiological stability
- **PC4** contributes $\sim 10\text{--}12\%$ $\rightarrow$ Strong respiratory component (paCO_2 , RBC-related oxygen transport)
- **PC119** contributes $\sim 6\text{--}8\%$ $\rightarrow$ Captures nuanced trajectory patterns (respiratory variability, CO_2 dynamics, discharge-related variables)
- Remaining top components (PC18, PC75, PC78, PC118, etc.) each contribute $2\text{--}4\%$ $\rightarrow$ Smaller but meaningful signals related to renal function, inflammation, and hemodynamics

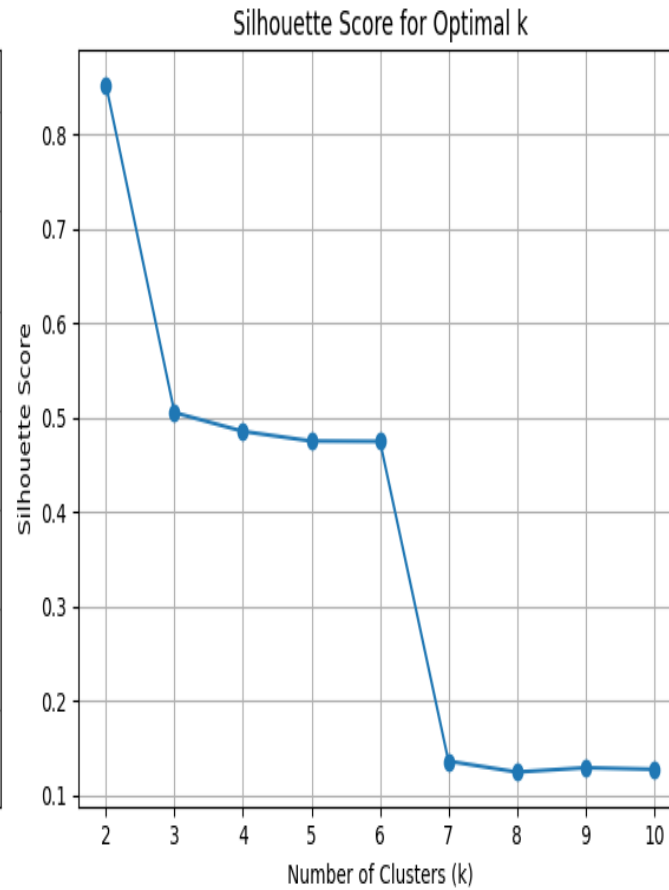
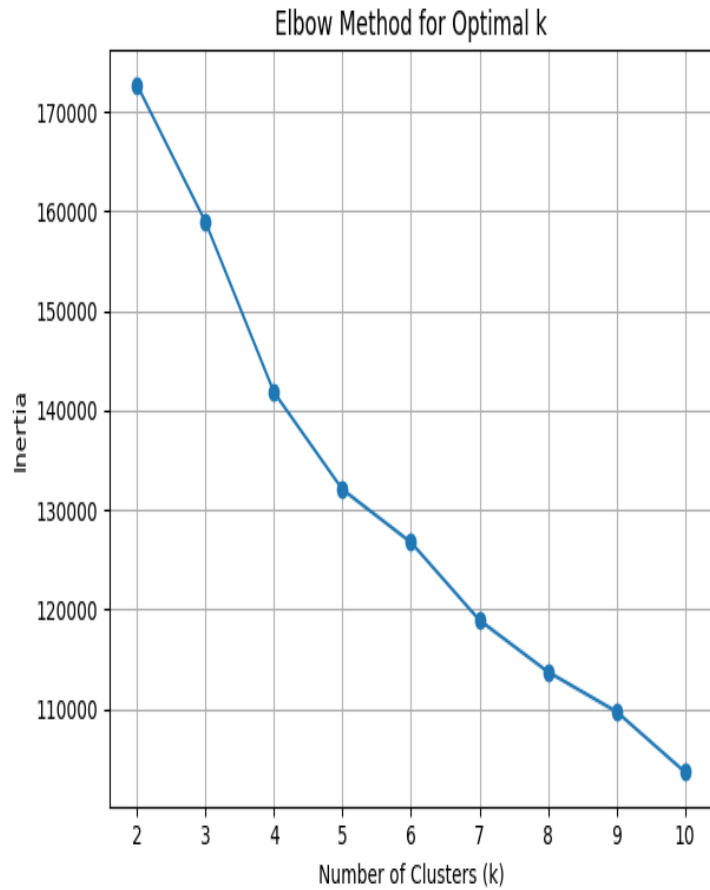


The background of the slide is a grayscale image of an ECG (heart rate) monitor. It shows several horizontal traces of electrical activity on a grid. The traces are slightly blurred and have a dark, muted color palette. The grid lines are visible, and there are some faint handwritten markings on the paper.

UNSUPERVISED MODELLING

- K-Means Clustering
- Hierarchical Clustering

K-MEANS CLUSTERING



Optimal $k = 2$

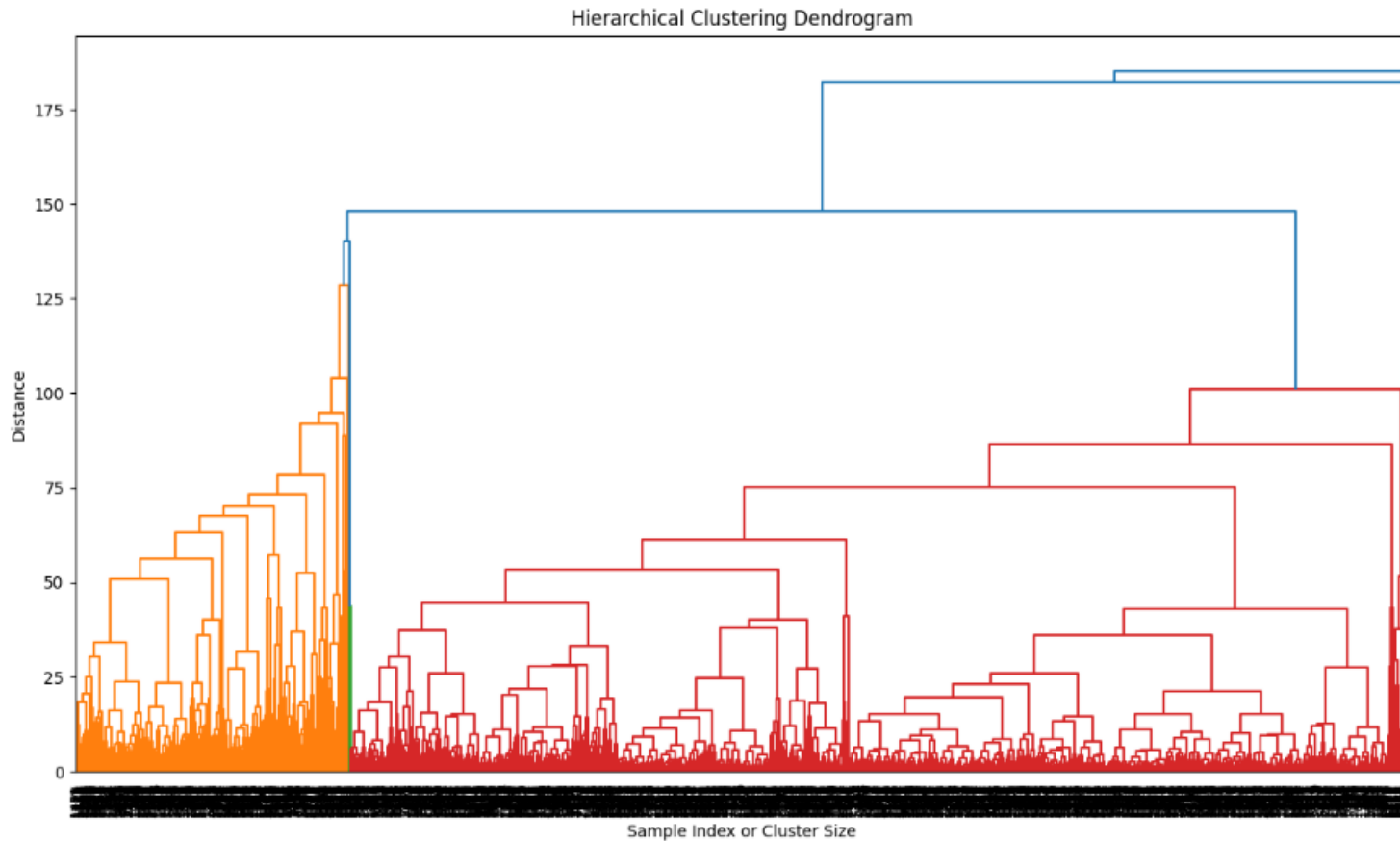
- Silhouette Score peaks at 0.85 → strong separation for $k = 2$
- Elbow Method shows diminishing returns after $k = 2$

- **Cluster Sizes:**
 - Cluster 0: 2,367 patients (99.66%)
 - Cluster 1: 8 patients (0.34%) → extreme imbalance
- **Vital Sign Differences (Means):**
 - *Respiration Rate*: 21.52 (Cluster 1) vs 19.57 (Cluster 0)
 - *CVP*: 11.72 (Cluster 1) vs 16.11 (Cluster 0)
 - *SpO₂*: 96.92% (Cluster 1) vs 95.96% (Cluster 0)
- **Interpretation:**
 - Cluster 1 shows higher respiratory drive + lower venous pressure
 - Suggests a physiologically distinct but rare subgroup

K-MEANS CLUSTERING

Metric	Cluster 0	Cluster 1
Number of Patients	2367	8
Heart Rate (mean)	83.78	79.27
Temperature (mean)	37.15	37.12
Oxygen Saturation (SpO ₂ mean)	95.96	96.92
Respiration Rate (mean)	19.57	21.52
CVP (mean)	16.11	11.72
End-tidal CO ₂ (mean)	29.91	30.00
Systemic Systolic BP (mean)	122.73	123.21
Systemic Diastolic BP (mean)	61.39	60.07
Systemic Mean BP (mean)	80.65	79.32
PA Mean Pressure (mean)	25.15	23.84
ICP (mean)	11.03	11.00

HIERARCHICAL CLUSTERING



Three distinct clusters emerged using Ward's method

Cluster Sizes:

- **Cluster 1:** 1,102 patients (~46%)
- **Cluster 2:** 842 patients (~35%)
- **Cluster 3:** 456 patients (~19%)

Mortality Rates:

- **Cluster 1:** ~20% mortality → highest-risk group
- **Cluster 2:** ~9% mortality → intermediate severity
- **Cluster 3:** 0% mortality → lowest-risk, physiologically stable

HIERARCHICAL CLUSTERING

- **Physiological Patterns:**
 - Cluster 1: elevated APACHE, high variability in vitals, more ventilation days
 - Cluster 2: moderate severity, moderate variability
 - Cluster 3: stable vitals, lower APACHE, minimal respiratory instability
- **Interpretation**
 - Hierarchical clustering captures **nested severity gradients** that K-Means missed
 - Clusters map onto **clinically meaningful phenotypes**
 - Suggests ICU patients follow **distinct physiological trajectories** rather than a single deterioration pathway

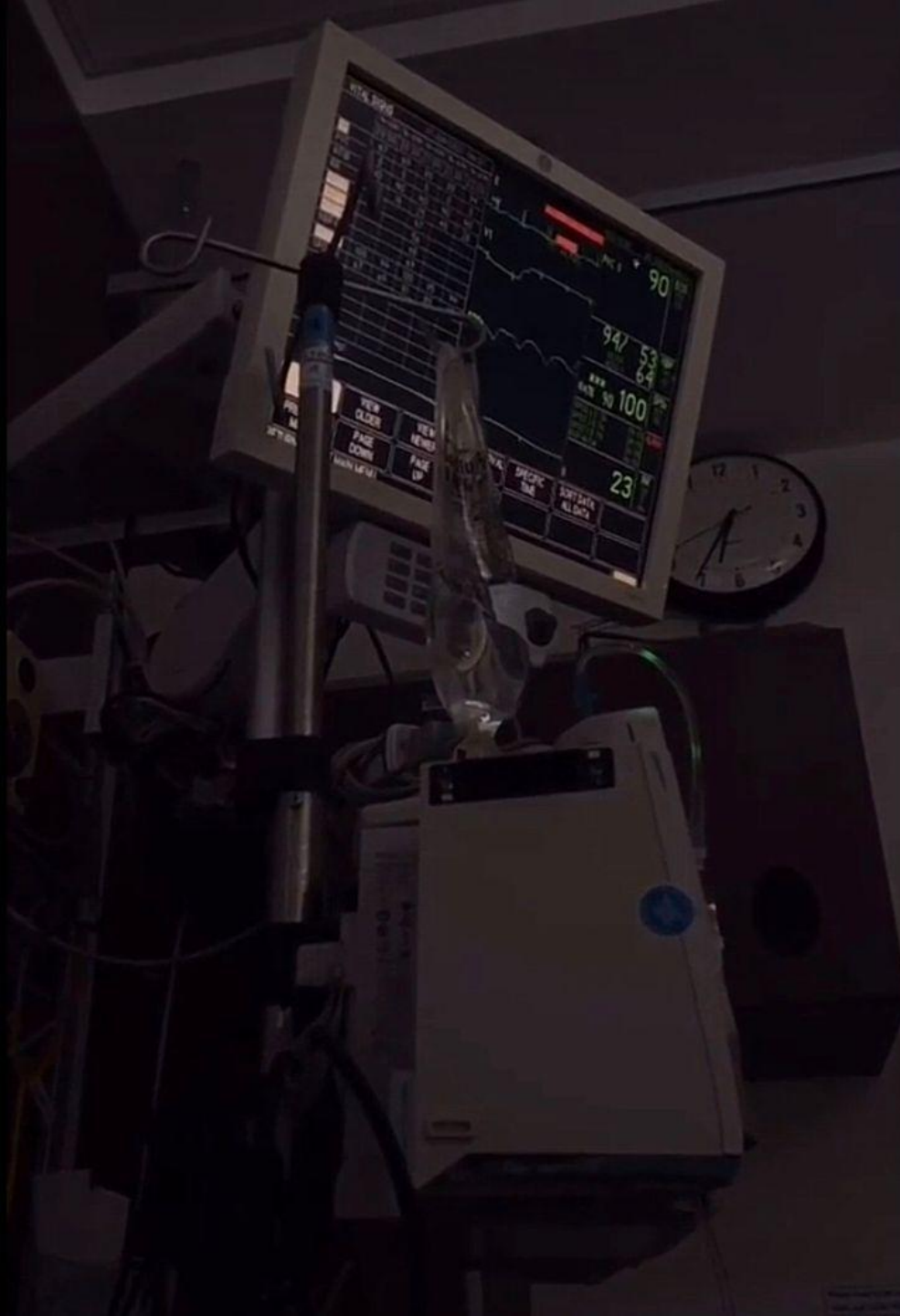
CLINICAL IMPLICATIONS



- **Early identification of high-risk patients**
 - LightGBM achieves >0.99 AUC, enabling reliable early risk stratification
 - PC1 and PC4 capture global severity + respiratory instability within the first 24 hours
- **Supports proactive interventions**
 - High-risk patients could receive earlier respiratory support, fluid management, or closer monitoring
 - Reduces time-to-intervention during deterioration windows
- **Improves resource allocation**
 - Helps prioritize ICU beds, ventilators, and staffing for patients with the highest predicted risk
 - Clusters reveal distinct physiological trajectories, enabling tailored monitoring intensity
- **Enhances clinical decision support**
 - Models can augment—not replace—clinical judgment
 - Provides continuous, data-driven risk estimates rather than static APACHE snapshot

LIMITATIONS

- PCA reduces interpretability
- Clustering imbalance
- No deep temporal modeling
- No external validation
- Potential dataset bias



FUTURE DIRECTIONS

- Integrate deep temporal models
- Improve interpretability of PCA-based models
- Refine clustering with advanced methods
- Enhance feature engineering

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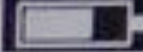
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